Fixed-Lag Correlated SMC^2: Scalable Online Bayesian Parameter Learning for Nonlinear State-Space Models

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Abstract

Sequential Monte Carlo squared (SMC^2) provides a principled framework for online Bayesian parameter learning in state-space models, but its reliance on full-history particle Markov chain Monte Carlo (PMCMC) rejuvenation renders it computationally intractable for long time series. We present three algorithmic innovations that reduce the per-rejuvenation cost from $O(T)$ to $O(L)$ while maintaining valid posterior inference: (i) a *fixed-lag checkpoint* scheme that truncates PMCMC replay to the most recent L observations, justified by exponential forgetting in the latent state; (ii) integration of *correlated* PMCMC (CPMMH) into the SMC^2 rejuvenation step, with a double-buffered noise management scheme for the θ-particle population; and (iii) a *canonical sorting* procedure that restores particle-index alignment after systematic resampling, enabling meaningful noise correlation across current and proposed paths. We demonstrate the method on a stochastic volatility model with mixture observation noise, where the inner filter is *Rao-Blackwellized* to reduce the effective sampling dimension from two to one—a design choice we argue is essential, not optional, for the outer SMC^2 and CPMMH machinery to function. At the scale of 1024 outer particles $\times$ 512 inner particles, the algorithm processes 3000 observations per window in $\sim$2 seconds on a single GPU. The algorithm achieves 6–8% CPMMH acceptance rates and accurate posterior tracking through parameter regime changes, where standard SMC^2 is computationally infeasible.

Keywords: Sequential Monte Carlo, particle MCMC, online parameter learning, correlated pseudo-marginal methods, Rao-Blackwellization, stochastic volatility, GPU computing

1 Introduction

State-space models (SSMs) are ubiquitous in time-series analysis, with applications ranging from financial econometrics to signal processing and target tracking. A persistent challenge is the simultaneous online estimation of static parameters θ and latent states $x_{1:t}$ as observations $y_{1:t}$ arrive sequentially. While sequential Monte Carlo (SMC) methods—particle filters—provide efficient online state estimation for fixed θ, learning θ itself requires a second layer of inference.

SMC^2 [Chopin et al., 2013] addresses this by embedding a particle filter within each θ-particle of an outer SMC sampler. As new data arrive, the outer weights $w_i \propto \hat{p}(y_{1:t} \,|\, \theta_i)$ are updated using the marginal likelihood estimates from each inner filter. When the outer effective sample size (ESS) drops below a threshold, the population is resampled and rejuvenated via particle MCMC (PMCMC; Andrieu et al. 2010).

The computational bottleneck is rejuvenation. Standard SMC^2 replays the *entire* observation history $y_{1:t}$ to compute the marginal likelihood under a proposed θ^*. After T observations with rejuvenation triggered every ΔT steps, the total cost scales as $O(T^2/\Delta T)$—quadratic in the series length. For high-frequency financial data, where T may reach 10^5–10^6 within a single trading session, this is prohibitive.

A second, more subtle problem afflicts PMCMC within SMC^2: the variance of the log-likelihood estimator $\log \hat{p}(y_{1:t} \mid \theta)$ grows with t, degrading Metropolis–Hastings acceptance rates. Deligiannidis et al. [2018] showed that correlating the random numbers between current and proposed paths dramatically reduces this variance for standalone PMCMC. However, integrating correlated PMCMC into the multi-particle SMC^2 framework introduces nontrivial bookkeeping challenges, particularly regarding noise buffer management and the preservation of correlation structure across resampling events.

Contributions. We introduce three modifications to SMC^2 that jointly make it practical for long, streaming time series:

1. **Fixed-lag checkpointing** (§3): PMCMC replay is truncated to the most recent L observations by checkpointing the inner particle filter state. We provide conditions under which the approximation error decays exponentially in L.

2. **Correlated PMCMC integration** (§4): We embed the correlated pseudo-marginal Metropolis–Hastings (CPMMH) algorithm of Deligiannidis et al. [2018] into SMC^2's rejuvenation step, with a double-buffered noise storage scheme that supports N_θ concurrent θ-particles with independent noise histories.

3. **Canonical sorting for correlation preservation** (§5): We introduce a bitonic sort on inner particles after systematic resampling that restores a canonical ordering by latent state value, enabling meaningful noise correlation between current and proposed particle populations.

4. **Rao-Blackwellized inner filter** (§7.2): We demonstrate that the inner particle filter must be Rao-Blackwellized—analytically marginalizing the conditionally Gaussian state—for SMC^2 with CPMMH rejuvenation to achieve viable acceptance rates. This reduces the effective sampling dimension exponentially, controlling the log-likelihood variance that governs the entire method's performance.

Together, these reduce the per-rejuvenation cost from $O(T \cdot N_x)$ to $O(L \cdot N_x)$ and the log-likelihood ratio variance from $O(T)$ to $O(L \cdot (1-\rho^2))$, where ρ is the noise correlation parameter.

2 Background

2.1 State-Space Models and Particle Filtering

Consider a general SSM with parameters θ, latent states x_t, and observations y_t:

$$x_t \mid x_{t-1}, \theta \sim f_\theta(x_t \mid x_{t-1}), \tag{1}$$

$$y_t \mid x_t, \theta \sim g_\theta(y_t \mid x_t). \tag{2}$$

A bootstrap particle filter (BPF) with N_x particles provides an unbiased estimate of the marginal likelihood:

$$\hat{p}(y_{1:T} \mid \theta) = \prod_{t=1}^{T} \left(\frac{1}{N_x} \sum_{i=1}^{N_x} w_t^{(i)} \right), \tag{3}$$

where $w_t^{(i)} = g_\theta(y_t \mid x_t^{(i)})$ are the unnormalized importance weights after propagation from the transition prior.

2.2 SMC^2

SMC^2 [Chopin et al., 2013] maintains a population of N_θ parameter particles $\{\theta_i\}_{i=1}^{N_\theta}$, each equipped with its own N_x-particle filter. The outer importance weights are the marginal likelihood estimates: $W_i^{(t)} \propto \hat{p}(y_{1:t} \,|\, \theta_i)$.

When the outer ESS drops below a threshold $\tau \cdot N_\theta$, a resample–rejuvenate step is performed. In the standard algorithm, rejuvenation proposes θ_i^* via a Metropolis–Hastings kernel and *re-runs the entire particle filter from* $t = 0$ under θ_i^* to obtain $\hat{p}(y_{1:t} \,|\, \theta_i^*)$.

The acceptance probability is:

$$\alpha = \min\left(1, \frac{\hat{p}(y_{1:t} \,|\, \theta^*)\, \pi(\theta^*)\, q(\theta \,|\, \theta^*)}{\hat{p}(y_{1:t} \,|\, \theta)\, \pi(\theta)\, q(\theta^* \,|\, \theta)}\right). \tag{4}$$

This is valid because $\hat{p}(y_{1:t} \,|\, \theta)$ is an unbiased estimator of the true marginal likelihood [Andrieu et al., 2010].

2.3 The Quadratic Cost Problem

Let rejuvenation be triggered at times $\{t_k\}$ with roughly uniform spacing ΔT. Each rejuvenation replays from $t = 0$:

$$\text{Total cost} = \sum_k t_k \cdot N_x \approx \frac{T^2}{2\Delta T} \cdot N_x. \tag{5}$$

For $T = 50{,}000$, $N_x = 512$, $\Delta T = 500$: total replay cost $\approx 2.5 \times 10^9$ particle–steps. This is the fundamental barrier to deploying SMC^2 on long time series.

2.4 Correlated PMCMC

Deligiannidis et al. [2018] observed that the variance of the log-likelihood ratio $\log \hat{p}(y_{1:T} \,|\, \theta^*) - \log \hat{p}(y_{1:T} \,|\, \theta)$ grows linearly in T when the random numbers driving both filters are independent. By correlating the uniform random variables u_t^* and u_t through:

$$u_t^* = \rho\, u_t + \sqrt{1 - \rho^2}\, v_t, \quad v_t \sim \mathcal{N}(0, 1), \tag{6}$$

the variance of the log-ratio is reduced by a factor of $(1 - \rho^2)$, dramatically improving acceptance rates for long series.

3 Fixed-Lag Checkpointing

3.1 Exponential Forgetting in the Inner Filter

The key observation enabling checkpoint truncation is that in models with contractive latent dynamics, the particle filter's dependence on initial conditions decays exponentially. For the stochastic volatility model considered here, the latent state $\tilde{z}_t$ follows an AR(1) process with coefficient $\rho < 1$, and the Kalman filter state (m_t, P_t) for the log-variance process satisfies:

$$\|P_t - P_\infty\| \leq C \cdot |\phi|^{2t} \|P_0 - P_\infty\|, \tag{7}$$

where $\phi = 1 - \theta(z_t)$ is the local mean-reversion speed and P_∞ is the stationary Riccati solution. For typical parameters ($\phi \approx 0.85$–0.95), the initial condition is forgotten within $L \approx 30$–50 steps to machine precision.

3.2 Checkpoint Scheme

We maintain a checkpoint of the full inner particle state $\mathcal{C} = \{(\tilde{z}^{(j)}, m^{(j)}, P^{(j)}, w^{(j)})\}_{j=1}^{N_x}$ at a designated time t_c, updated every L observations. During PMCMC rejuvenation, instead of replaying from $t = 0$, we:

1. Initialize the proposed filter from the checkpoint $\mathcal{C}$ at t_c.
2. Replay only observations $y_{t_c+1}, \ldots, y_t$ under the proposed θ^*.
3. Compare the partial log-likelihoods: $\sum_{s=t_c+1}^{t} \log \hat{p}(y_s \mid y_{1:s-1}, \theta^*)$ vs. $\sum_{s=t_c+1}^{t} \log \hat{p}(y_s \mid y_{1:s-1}, \theta)$.

Proposition 1 (Checkpoint Bias Bound)**.** *Let $\hat{\alpha}_L$ be the acceptance probability using the fixed-lag checkpoint at lag L, and α the exact acceptance probability with full replay. Under the exponential forgetting condition* (7)*, the bias satisfies:*

$$|\mathbb{E}[\hat{\alpha}_L] - \mathbb{E}[\alpha]| \leq K \cdot |\phi|^{2L}, \tag{8}$$

for a constant K depending on N_x and the observation model. For $\phi = 0.9$ and $L = 50$, this gives $|\phi|^{100} \approx 2.7 \times 10^{-5}$.

3.3 Checkpoint Maintenance Under Outer Resampling

When the outer SMC^2 population is resampled, the ancestor indices $a_i \in \{1, \ldots, N_\theta\}$ determine which θ-particle each position inherits. The checkpoint arrays must be reindexed accordingly: $\mathcal{C}_i^{\text{new}} \leftarrow \mathcal{C}_{a_i}^{\text{old}}$. We implement this via a scatter kernel followed by a pointer swap between primary and scratch buffers, avoiding any device-to-device memory copies.

4 Correlated PMCMC in SMC^2

4.1 Noise Buffer Architecture

Each θ-particle i requires a history of driving noise for replay:

- $\varepsilon^z_{i,t,j}$: noise for propagating latent state $\tilde{z}$ (particle j at time t),
- $\varepsilon^u_{i,t}$: noise for the systematic resampling offset u_0 at time t.

We store these in circular buffers of capacity $C \geq 2L$ with modular time indexing: $\text{slot}(t) = t \bmod C$. The memory layout is $\varepsilon^z[i][t \bmod C][j]$ for coalesced GPU access.

Two copies of each buffer are maintained ("ping-pong"): $\varepsilon^{(0)}$ (current) and $\varepsilon^{(1)}$ (proposed). During CPMMH, the proposed noise is generated as:

$$\varepsilon^*_{i,t,j} = \rho_{\text{cpmmh}} \cdot \varepsilon_{i,t,j} + \sqrt{1 - \rho^2_{\text{cpmmh}}} \cdot \nu_{i,t,j}, \quad \nu_{i,t,j} \sim \mathcal{N}(0, 1). \tag{9}$$

Upon acceptance, the proposed buffer becomes current (pointer swap); upon rejection, the proposed buffer is discarded.

4.2 Integration with SMC^2 Rejuvenation

Algorithm 1 details the rejuvenation step. Each of K CPMMH iterations runs across all N_θ particles in parallel (one GPU block per θ-particle). The accept/reject decision is per-particle, so the population maintains diversity through independent mixing.

Algorithm 1 CPMMH Rejuvenation within SMC^2

Require: Particles $\{\theta_i, \varepsilon_i^{(b)}\}_{i=1}^{N_\theta}$, checkpoint $\mathcal{C}$, observations $y_{t_c+1:t}$

for $k = 1, \ldots, K$ **do**
  for $i = 1, \ldots, N_\theta$ **in parallel do**
    Propose $\theta_i^* \sim q(\cdot \mid \theta_i)$
    Generate correlated noise $\varepsilon_i^{(1-b)}$ via (9)
    Run particle filter from $\mathcal{C}_i$ under $(\theta_i^*, \varepsilon_i^{(1-b)})$
    Compute log-likelihood $\ell_i^* = \sum_{s=t_c+1}^{t} \log \hat{p}(y_s \mid \theta_i^*, \varepsilon_i^{(1-b)})$
    $\alpha_i \leftarrow \min(1,\ \exp(\ell_i^* - \ell_i + \log \pi(\theta_i^*) - \log \pi(\theta_i)))$
    if $U_i < \alpha_i$ **then**
      $\theta_i \leftarrow \theta_i^*, \quad \ell_i \leftarrow \ell_i^*$
      Commit: swap noise buffers for particle i
    end if
  end for
end for

5 Canonical Sorting for Correlation Preservation

5.1 The Resampling Correlation Problem

Systematic resampling within the inner particle filter is essential for maintaining ESS but destructive to noise correlation. Consider two filter runs (current and proposed) at time t. Before resampling, particle index j in both runs was propagated using $\varepsilon_{t,j}$, so correlating $\varepsilon_{t,j}$ and $\varepsilon_{t,j}^*$ produces correlated latent states. After resampling, particle j in the current run originated from ancestor a_j, while in the proposed run it came from a_j^*. Since $a_j \neq a_j^*$ in general, the index-based noise correlation is meaningless—correlated noise at index j drives unrelated particles.

5.2 Sorting by Latent State

Our solution is to *sort* inner particles by their latent state value $\tilde{z}^{(j)}$ after resampling. This establishes a canonical ordering where index j corresponds to the j-th quantile of the empirical $\tilde{z}$ distribution. Since both current and proposed paths share similar (correlated) $\tilde{z}$ distributions, the j-th quantile particle in both populations represents a similar latent state. Correlating noise at index j then produces correlated state transitions.

Concretely, every K_{sort} timesteps (after systematic resampling):

1. Sort inner particles by $\tilde{z}^{(j)}$ ascending, reindexing the co-located Kalman state $(m^{(j)}, P^{(j)})$ accordingly.
2. Apply the same permutation to the inner weight array (all weights are equal post-resample, so this is a no-op in practice).

The sort is implemented as a bitonic sort in GPU shared memory. For $N_x = 512$, this requires $\frac{1}{2} N_x \log^2 N_x \approx 20{,}000$ comparisons, completing in $< 5\,\mu$s. The cost is negligible relative to the OCSN likelihood evaluations.

Remark 1. *The sorting frequency K_{sort} controls a bias–variance tradeoff. Sorting every tick ($K_{sort} = 1$) maximizes correlation preservation but adds overhead. In practice, $K_{sort} = 4$–7 maintains sufficient correlation, as the particle ordering drifts slowly between resampling events.*

5.3 Why Sorting Works: Quantile Coupling Argument

The effectiveness of sorting rests on a quantile coupling. Let F and F^* denote the empirical CDFs of $\tilde{z}$ under the current and proposed parameters after resampling. If $\theta^* \approx \theta$ (as ensured by a local random walk proposal), then $\|F - F^*\|_\infty$ is small. The sorted particles at index j satisfy:

$$|\tilde{z}^{(j)} - \tilde{z}^{*(j)}| \leq \|F - F^*\|_\infty \cdot \mathrm{range}(\tilde{z}) + O(N_x^{-1/2}). \tag{10}$$

Thus, correlated noise $\varepsilon_j^* = \rho\,\varepsilon_j + \sqrt{1-\rho^2}\,\nu_j$ applied to nearly identical initial states produces highly correlated next states, achieving the variance reduction intended by (6).

Without sorting, the mapping between indices is random, and the effective correlation is:

$$\rho_{\mathrm{eff}} \approx \rho \cdot \Pr(a_j = a_j^*) \ll \rho, \tag{11}$$

since independent resampling draws agree only with probability $O(1/\mathrm{ESS})$ per index.

6 Computational Complexity

Table 1 summarizes the cost comparison.

Table 1: Per-rejuvenation cost comparison.

Method	Cost per rejuvenation	Log-ratio variance
Standard SMC^2	$O(T \cdot N_x)$	$O(T)$
Fixed-lag only	$O(L \cdot N_x)$	$O(L)$
CPMMH (no sort)	$O(L \cdot N_x)$	$O(L)$ (correlation lost)
FL-CPMMH + sort	$O(L \cdot N_x)$	$O(L \cdot (1-\rho^2))$

With $L = 50$, $N_x = 512$, $\rho = 0.99$, and $K = 3$ rejuvenation steps, each rejuvenation event requires $3 \times 50 \times 512 = 76{,}800$ particle–steps. At $T = 3{,}000$ with one rejuvenation, the total is $3{,}000 \times 512 + 76{,}800 \approx 1.6 \times 10^6$ particle–steps, compared to $3{,}000 \times 512 \times 2 \approx 3 \times 10^6$ for standard SMC^2 (and growing quadratically with longer windows).

The log-ratio variance reduction from correlated noise is a factor of $1-\rho^2 = 1-0.99^2 \approx 0.02$, yielding a $50\times$ reduction. This translates to CPMMH acceptance rates of 6–8% versus effectively 0% without correlation.

7 Application: Stochastic Volatility with OCSN Noise

7.1 Model Specification

We consider a stochastic volatility model for high-frequency asset returns. The latent log-variance h_t evolves as a mean-reverting process driven by a latent regime variable z_t:

$$\tilde{z}_t = \rho\,\tilde{z}_{t-1} + \sigma_z\,\varepsilon_t^z, \quad z_t = z_c\,(1 + \tanh(\tilde{z}_t)), \tag{12}$$

$$h_t = \phi(z_t)\,h_{t-1} + \theta(z_t)\,\mu(z_t) + \sigma_h(z_t)\,\varepsilon_t^h, \tag{13}$$

$$y_t = \exp(h_t/2)\,\eta_t, \quad \eta_t \sim \mathrm{OCSN}_{10}, \tag{14}$$

where $\phi(z) = 1 - \theta(z)$, and $\theta(\cdot)$, $\mu(\cdot)$, $\sigma_h(\cdot)$ are parameterized as saturating curves: $f(z) = f_{\mathrm{base}} + f_{\mathrm{scale}}\,(1 - e^{-f_{\mathrm{rate}} \cdot z})$.

The OCSN (Order-Statistics-based Contaminated Normal) observation distribution is a 10-component Gaussian mixture fitted to empirical microstructure noise, providing robustness to the heavy tails and multimodality characteristic of high-frequency returns.

7.2 Rao-Blackwellization of the Inner Filter

A critical architectural choice is the use of a *Rao-Blackwellized* particle filter (RBPF) for the inner SMC layer. The model (12)–(14) admits a conditional Gaussian structure: given a trajectory of the regime variable $z_{1:t}$, the log-variance process h_t evolves as a linear Gaussian state-space model with known (time-varying) parameters $\phi(z_t)$, $\mu(z_t)$, and $\sigma_h(z_t)$. The RBPF exploits this by maintaining particles only over $\tilde{z}_t$ and tracking the conditional distribution $p(h_t \mid \tilde{z}_{1:t}, y_{1:t})$ analytically via Kalman filtering. Each inner particle carries a tuple $(\tilde{z}_t^{(j)}, m_t^{(j)}, P_t^{(j)})$: the sampled regime state and the conditional mean and variance of h_t.

This is not merely an efficiency optimization—it is *essential* for SMC^2 to function at all. The argument has three layers:

1. Variance of the log-likelihood estimator. The quality of SMC^2 hinges on the variance of the inner filter's marginal likelihood estimate $\hat{p}(y_{1:T} \mid \theta)$. For CPMMH to achieve nonzero acceptance rates, the variance of $\log \hat{p}(y_{1:T} \mid \theta)$ must be controlled. With a standard BPF sampling both $\tilde{z}_t$ and h_t as particles, the effective state dimension is 2. It is well established [Snyder et al., 2008] that the required number of particles to maintain a given ESS grows exponentially with the state dimension d:

$$N_x \gtrsim \exp\big(c \cdot d \cdot \mathrm{Var}[\log w_t]\big). \tag{15}$$

Rao-Blackwellization reduces the effective sampling dimension from $d = 2$ (sampling $\tilde{z}_t$ and h_t jointly) to $d = 1$ (sampling only $\tilde{z}_t$), yielding an *exponential* reduction in the number of particles required. For $N_x = 512$, the univariate RBPF provides adequate ESS; a joint BPF would require orders of magnitude more particles for equivalent performance.

2. Compounding through the CPMMH chain. The CPMMH replay replays L timesteps to compute the likelihood ratio. At each replayed tick, the inner filter performs a full resample–propagate–weight cycle. Per-tick weight variance compounds multiplicatively in the log-likelihood:

$$\mathrm{Var}\Big[\log \hat{p}(y_{t_c+1:t} \mid \theta)\Big] = \sum_{s=t_c+1}^{t} \mathrm{Var}\big[\log \hat{p}(y_s \mid y_{1:s-1}, \theta)\big]. \tag{16}$$

With the BPF, each summand is larger (due to higher-dimensional sampling), so the cumulative variance over L steps grows faster, degrading the CPMMH log-ratio signal. With the RBPF, the per-tick variance is smaller because the Kalman filter provides the exact conditional likelihood $p(y_t \mid \tilde{z}_t^{(j)}, m_{t-1}^{(j)}, P_{t-1}^{(j)})$ rather than a Monte Carlo approximation. This is the difference between 6–8% CPMMH acceptance rates (functional) and $< 1\%$ (degenerate).

3. Checkpoint fidelity. The fixed-lag checkpoint stores $(\tilde{z}^{(j)}, m^{(j)}, P^{(j)})$ per inner particle. Because m and P are *exact* conditional moments (not particle approximations), the checkpoint fully captures the filter state with no information loss beyond the particle approximation of the $\tilde{z}$ marginal. A standard BPF checkpoint would store $(\tilde{z}^{(j)}, h^{(j)})$—point samples of h rather than its full conditional distribution—losing the uncertainty quantification that the Kalman filter provides. Upon replay from checkpoint, this lost information manifests as additional variance in the likelihood estimate, further degrading CPMMH acceptance.

Remark 2. *The RBPF structure also enables the OCSN mixture observation model at no additional particle cost. The 10-component mixture likelihood is evaluated analytically given $(m^{(j)}, P^{(j)})$ via Gaussian–Gaussian convolution per component. A standard BPF would require either discretizing the mixture (losing accuracy) or additional importance sampling within each component (increasing cost).*

7.3 Parameter Learning Configuration

The full model has 8 parameters $\theta = (\rho, \sigma_{\text{total}}, r_{\text{split}}, \mu_{\text{base}}, \mu_{\text{scale}}, \mu_{\text{rate}}, \sigma_{\text{scale}}, \sigma_{\text{rate}})$. We demonstrate a 4-parameter "fast mode" where the curve shape parameters $(\mu_{\text{scale}}, \mu_{\text{rate}}, \sigma_{\text{scale}}, \sigma_{\text{rate}})$ are fixed, learning only $(\rho, \sigma_{\text{total}}, r_{\text{split}}, \mu_{\text{base}})$. This avoids a structural ridge in the likelihood surface: over the narrow range of z visited within a single window, the nonlinear curve $\mu_{\text{base}} + \mu_{\text{scale}}(1 - e^{-\mu_{\text{rate}} \cdot z})$ is indistinguishable from a linear approximation, creating a manifold of degenerate solutions.

7.4 GPU Implementation

The algorithm maps naturally to GPU architecture:

- Each θ-particle is a CUDA block (N_θ blocks).
- Each inner particle is a thread (N_x threads per block).
- Resampling, sorting, and ESS computation use shared-memory reductions.
- Noise buffers reside in global memory with coalesced access patterns.
- CPMMH rejuvenation re-uses the same block structure, replaying the particle filter within each block for the proposed θ^*.

At $N_\theta = 1024$, $N_x = 512$, the algorithm processes a 3000-tick window in $\sim$2.2 seconds on an NVIDIA RTX GPU under Windows (WDDM), including all resampling and rejuvenation overhead.

8 Experimental Results

8.1 Regime-Switching DGP

We validate on a synthetic data-generating process with a parameter shift at $t = 6000$: μ_{base} jumps from -1.0 to $+1.0$. The algorithm processes overlapping windows of $W = 3000$ ticks with stride 1000. Table 2 shows representative results in 4-parameter fast mode.

Table 2: Parameter recovery in 4-parameter mode (post-shift).

Parameter	True	Estimate	Std	Err%	z-score
ρ	0.950	0.943	0.010	-0.8%	0.75
σ_{total}	0.180	0.193	0.010	$+7.2\%$	1.31
r_{split}	0.555	0.559	0.032	$+0.7\%$	0.12
μ_{base}	1.000	0.948	0.100	-5.2%	0.52

All four parameters are recovered within 2σ of the truth, with μ_{base} correctly tracking the post-shift value. The algorithm detects the regime change within 1–2 windows (1000–2000 ticks) of the true shift point.

8.2 Ablation: Effect of CPMMH Correlation

Without the canonical sorting procedure, CPMMH acceptance rates drop from 6–8% to $< 1\%$, and the algorithm fails to maintain posterior diversity after resampling. Without correlated noise entirely (standard PMMH within SMC^2), acceptance rates are effectively zero for replay windows exceeding $\sim$20 ticks, rendering rejuvenation ineffective.

9 Discussion

Relationship to existing work. Fixed-lag approximations are well-established for state smoothing [Kitagawa and Sato, 2001], but their application to bounding PMCMC replay cost within SMC^2 appears to be novel. The CPMMH algorithm of Deligiannidis et al. [2018] was developed for standalone parameter estimation; embedding it in the multi-particle SMC^2 framework with per-particle noise management is a nontrivial extension. The sorting trick for preserving noise correlation across resampling has no direct precedent in the literature to our knowledge; the closest related work involves ancestor-tracking in conditional SMC [Lindsten et al., 2014], which addresses a different technical problem. Rao-Blackwellized particle filters [Doucet et al., 2000] are well-known, but their role as a *structural prerequisite* for SMC^2 with CPMMH—rather than a mere efficiency improvement—has not been explicitly articulated. Our analysis shows that without Rao-Blackwellization, the log-likelihood variance renders CPMMH acceptance rates effectively zero at the particle budgets feasible on current hardware.

Limitations. The fixed-lag approximation introduces a bias that scales as $|\phi|^{2L}$. For models with near-unit-root dynamics ($\rho \to 1$), larger L is required, increasing rejuvenation cost. The sorting trick assumes that the latent state is low-dimensional (univariate in our case); for multivariate latent states, defining a canonical ordering is less straightforward, though space-filling curve indices (e.g., Morton codes) could serve as a surrogate.

Observation model sensitivity. The OCSN mixture observation model, while robust to outliers in production, reduces Fisher information per observation relative to a Gaussian model. This softens the likelihood surface and slows parameter learning. For simulation-based validation, a Gaussian observation model may be preferable, with the OCSN model reserved for deployment on real market data.

10 Conclusion

We have presented a computationally tractable variant of SMC^2 for online Bayesian parameter learning in nonlinear state-space models. The combination of fixed-lag checkpointing, correlated pseudo-marginal methods with double-buffered noise management, and canonical sorting for correlation preservation reduces the asymptotic cost from $O(T^2)$ to $O(T \cdot L)$ while maintaining valid approximate posterior inference. The method achieves real-time performance on GPU hardware for stochastic volatility models at high-frequency timescales.

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